PROJECT 3: SQL DATA ANALYSIS

DecodeLabs Industrial Training | Batch 2026

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Tool: MySQL Workbench 8.0 | Database: decodelabs | Table: sales | Records: 1,200

Overview

This report presents six SQL queries run against a 1,200-row e-commerce sales dataset loaded into MySQL. Each query targets a specific business question using SELECT, WHERE, ORDER BY, GROUP BY, COUNT, SUM, and AVG. The results are shown below with the query code and output screenshot.

Query 1: View Data Sample

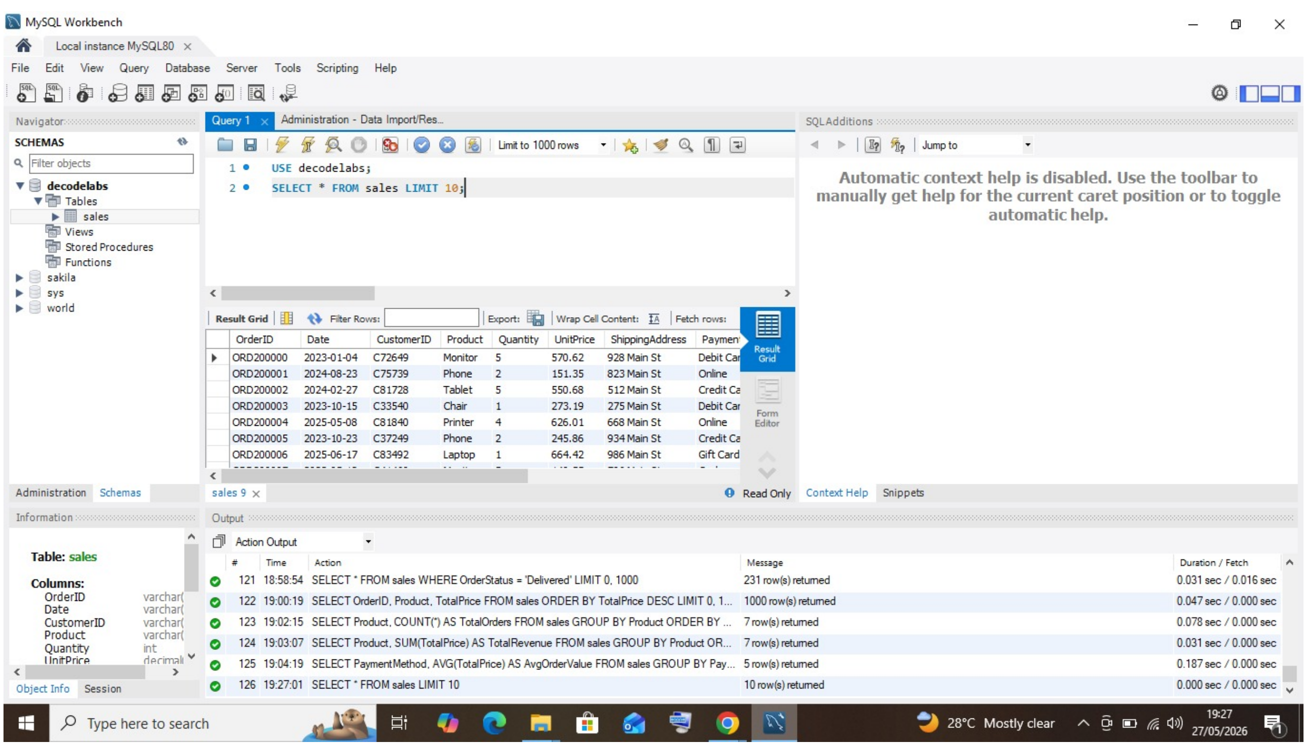
Purpose: Retrieve the first 10 rows to confirm the dataset loaded correctly and inspect column structure.

SQL Code:

```
USE decodelabs;
SELECT * FROM sales LIMIT 10;
```

Result:

10 rows returned. The table contains all 14 columns including OrderID, Date, Product, Quantity, UnitPrice, and TotalPrice.



Query 2: Filter Delivered Orders (WHERE)

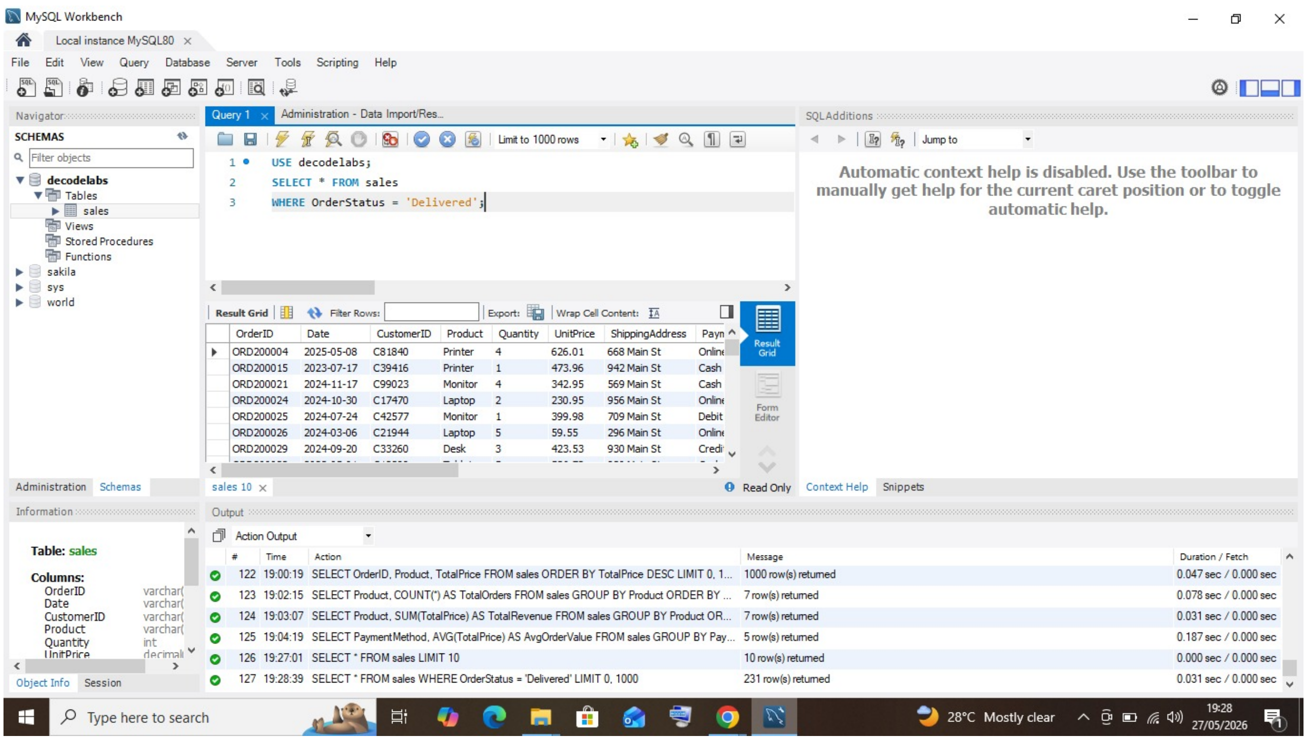
Purpose: Isolate all orders where the status is Delivered to understand fulfillment volume.

SQL Code:

```
USE decodelabs;
SELECT * FROM sales
WHERE OrderStatus = 'Delivered';
```

Result:

231 rows returned out of 1,200 total orders. This means roughly 19% of all orders were successfully delivered.



Query 3: Sort Orders by Total Price (ORDER BY)

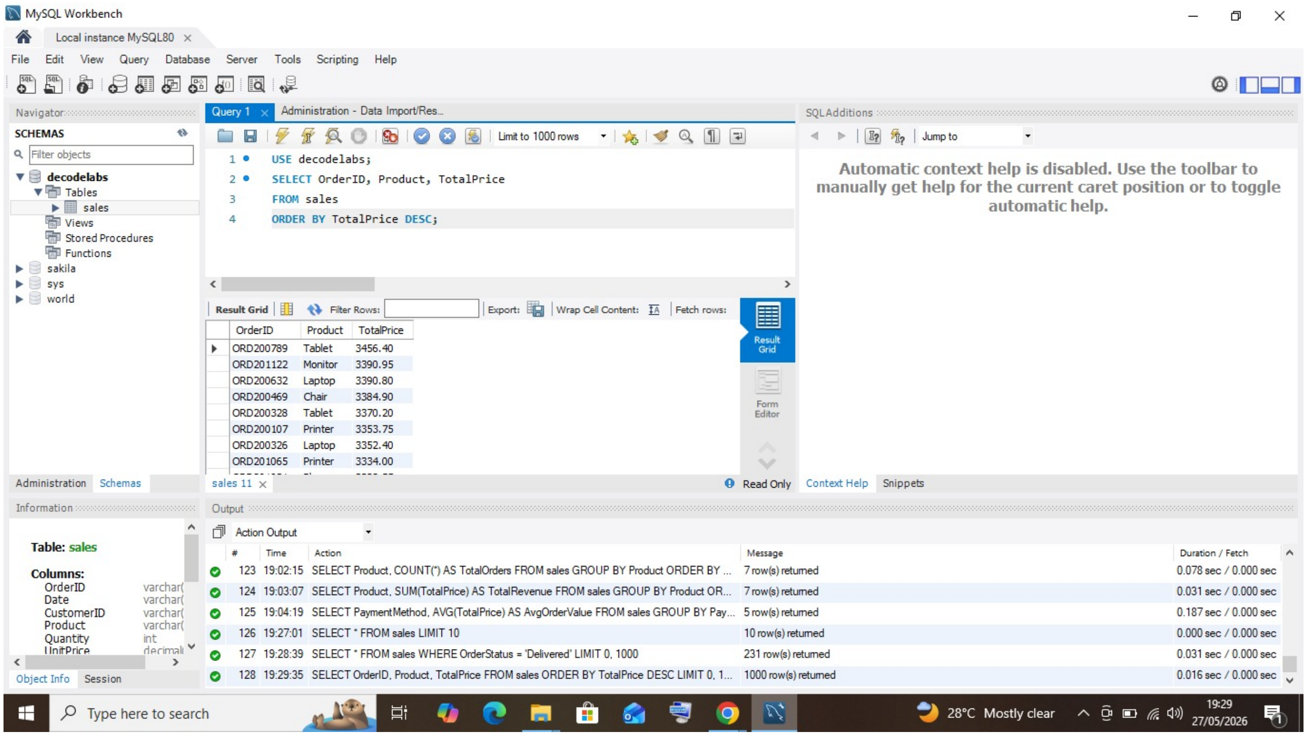
Purpose: Rank all orders from highest to lowest value to identify the most valuable transactions.

SQL Code:

```
USE decodelabs;
SELECT OrderID, Product, TotalPrice
FROM sales
ORDER BY TotalPrice DESC;
```

Result:

1,000 rows returned. The highest-value order was a Tablet (ORD200789) at 3,456.40. Tablets and Monitors dominate the top of the list.



Query 4: Count Orders per Product (COUNT + GROUP BY)

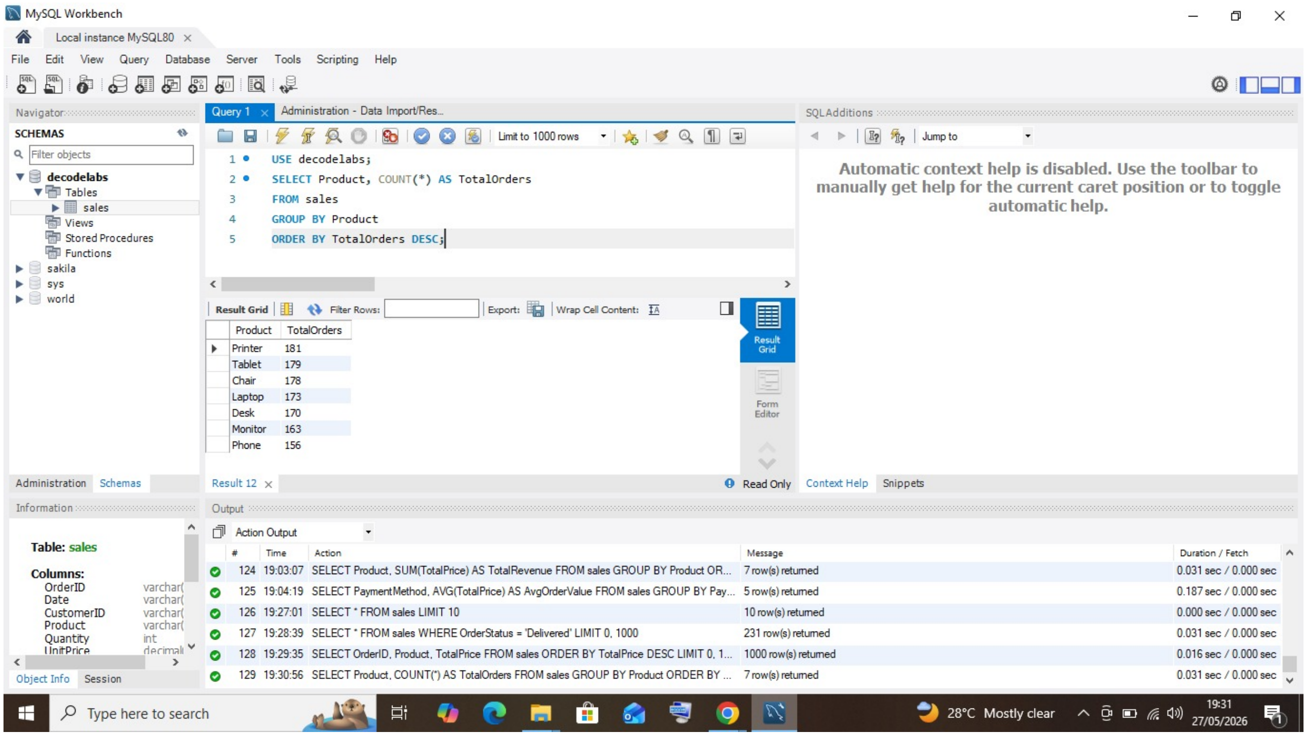
Purpose: Count how many orders exist for each product type to identify the most ordered product.

SQL Code:

```
USE decodelabs;
SELECT Product, COUNT(*) AS TotalOrders
FROM sales
GROUP BY Product
ORDER BY TotalOrders DESC;
```

Result:

7 product categories found. Printer leads with 181 orders, followed by Tablet (179) and Chair (178). Phone has the fewest at 156.



Query 5: Total Revenue per Product (SUM + GROUP BY)

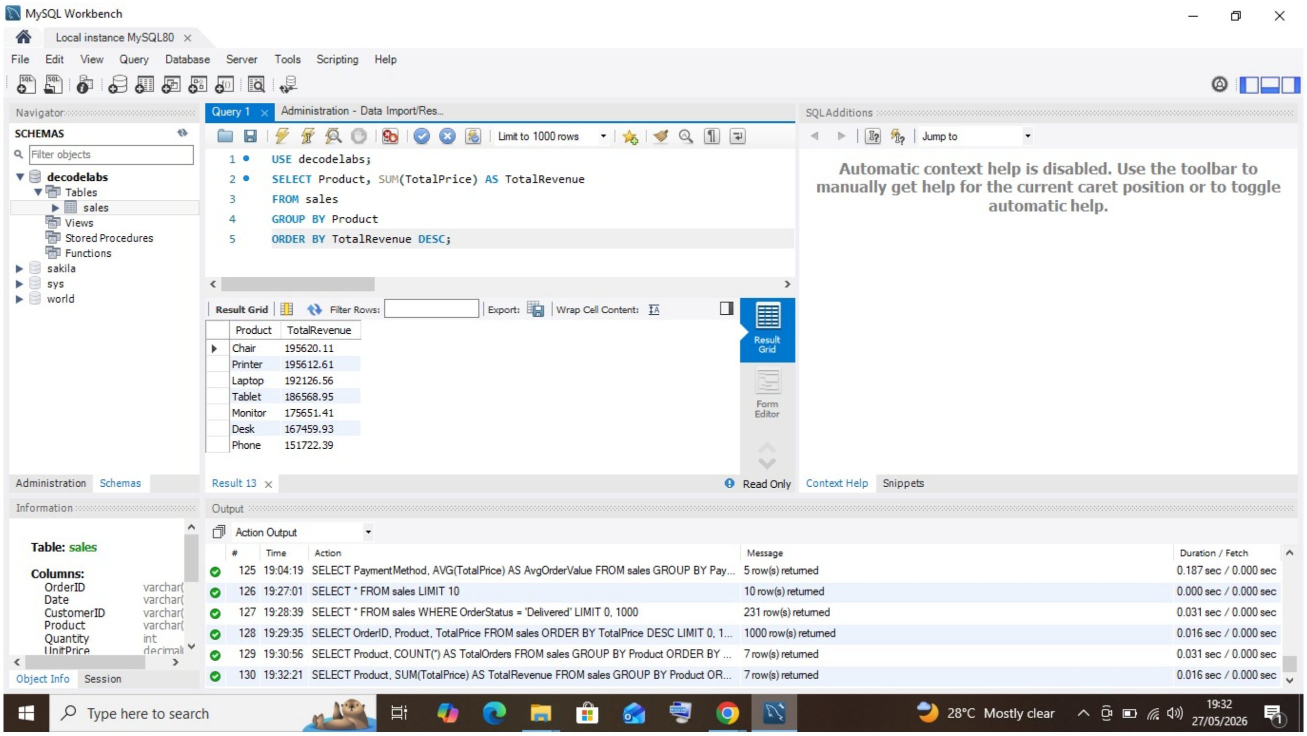
Purpose: Calculate total revenue generated by each product to identify the highest earning category.

SQL Code:

```
USE decodelabs;
SELECT Product, SUM(TotalPrice) AS TotalRevenue
FROM sales
GROUP BY Product
ORDER BY TotalRevenue DESC;
```

Result:

Chair generated the most revenue at 195,620.11, closely followed by Printer at 195,612.61. Phone had the lowest revenue at 151,722.39.



Query 6: Average Order Value by Payment Method (AVG + GROUP BY)

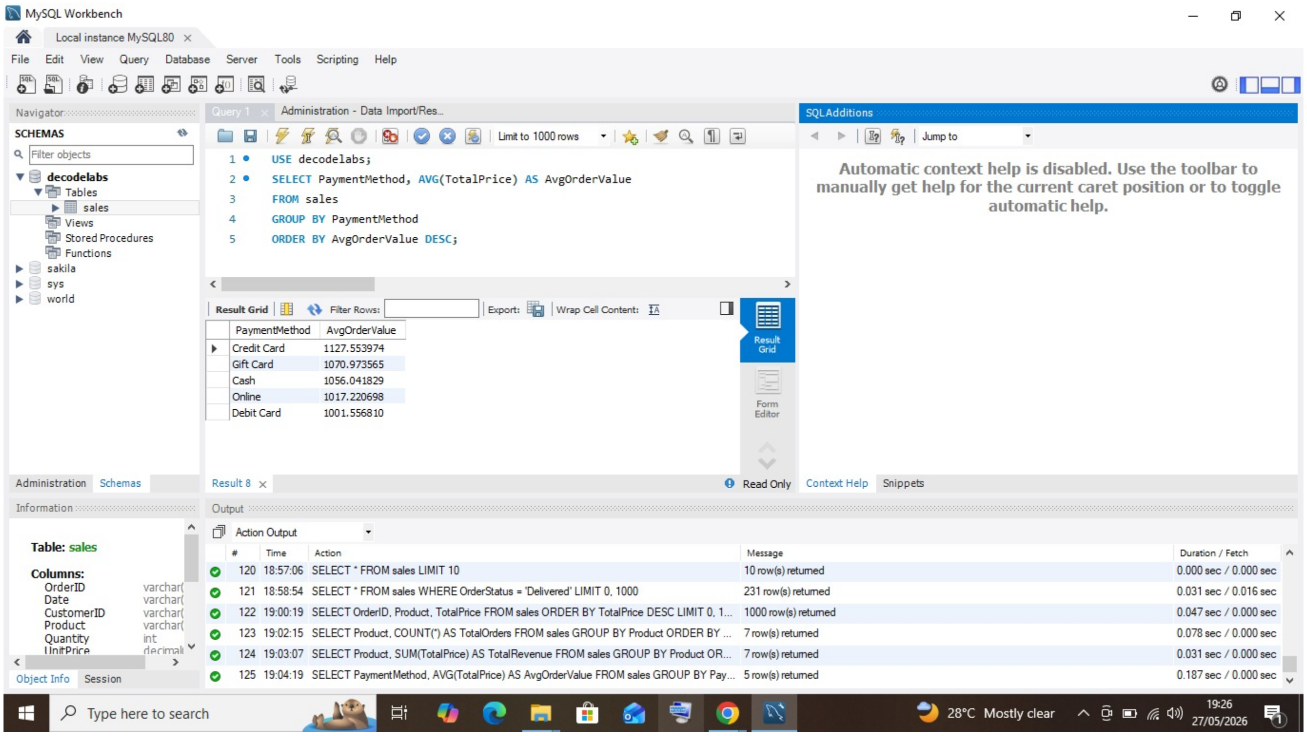
Purpose: Calculate the average order value for each payment method to understand customer spending by payment type.

SQL Code:

```
USE decodelabs;
SELECT PaymentMethod, AVG(TotalPrice) AS AvgOrderValue
FROM sales
GROUP BY PaymentMethod
ORDER BY AvgOrderValue DESC;
```

Result:

5 payment methods found. Credit Card customers spend the most on average at 1,127.55 per order. Debit Card customers have the lowest average at 1,001.56.



Summary of Findings

The following business insights emerged from the six SQL queries:

1. The dataset contains 1,200 orders across 7 product categories and 5 payment methods.

2. Only 231 orders (19%) have a Delivered status, which may indicate a high rate of returns, cancellations, or pending orders worth investigating.

3. Tablets appear among the highest-value individual orders, reaching up to 3,456.40 per transaction.

4. Printers are ordered most frequently (181 orders), but Chairs generate the most total revenue (195,620.11). This gap suggests Chairs carry a higher unit price despite fewer orders.

5. Credit Card is the highest-value payment method by average order, while Debit Card customers spend the least on average. This pattern could inform payment-specific promotions.

6. Phone is both the least ordered product and the lowest revenue generator, making it a candidate for marketing review.